Exp No : 12 End to end Communication at transport layer

1/10/26

Aim :

To implement an echo - client Server using TCP/UDP and implement a chat client - sever using Socket programming

(a) echo - client Server using TCP/UDP a Socket is an endpoint used for Communication b/w two Service. In an echo program, the client sends a message to server and the Server sends the same message back to client

Procedure :

(1) Create sower program and client program

(2) Import the required socket library

(3) In server, Creates a socket and bind it to an Ip address and Port

(4) Make the server listen to client request

(5) start client and Connect it to server

(6) enter msg from client

(7) Send msg to server

(8) The server receives the msg and sends msg back

(9) The client receives and display the echoed msg

⑩ close socket connection

⑥ client server lying socket program.

① Create server and client program lying socket programming
② Start server and bind at to an Ip address and port
③ Make server wait for client communication.
④ Start client and connect it to server
⑤ Enter msg in client
⑥ send msg to server
⑦ The server receives and display msg
⑧ The server sends a reply back to client
⑨ Continue sending and receiving msg to perform a chat.
⑩ close the connection after the chart is completed

output :

Enter message (or 'quit' to stop): [Server] Running on 127.0.
1200 hi
[SERVER] Received from ('127.0.0.1', 49870): hi
echo from server : hi
enter message (or 'quit' to stop): hello
[SERVER] Received from ('127.0.0.1', 49670): hello
echo from server : hello
enter message (or 'quit to stop ): quit
[client] (closing Connection.. [server] Client
('127.0.0.1', 4970 ended the Connection.
[client] stopped. [SERVER] stopped.

result :

The end to end Communication was successfully impl
using TCP/UDP sockets